Dg categories (Ben Gunnarsson)

History: 1990 Bondal, Kapranov "Enhanced triangulated categories"
2002 Drinfeld "Dg quotients of dg categories" "Q: what is dg cat from?"
2005 Tabuada Model structure on dg cat
2006 Toen "Homotopy theory of dg ... Morita ... ∞-Br cat."
Lurie "Higher alg": dg alg. from $(\infty,2)$-category.

Why dg categories?
- remember chain complexes
- curves functorial
- dg quotients
- limits, colimits (gluing dg cat).
- correct target for categorical invariants

Ex: $\mathbb{P}^1 = \mathrm{colim}\begin{pmatrix} \mathbb{A}^1 & \to & \mathbb{A}^1 \\ \downarrow & & \downarrow \\ \mathbb{A}^1 & \to & \mathbb{P}^1 \end{pmatrix}$ $\quad QCh(\mathbb{P}^1) = \lim\begin{pmatrix} \cdots & \cdots \end{pmatrix}$

Def: An $(\infty,1)$-category is stable if
(1) $0 \in C$
(2) every morphism has fiber + cofiber (i.e. kernel + cokernel)
(3) fiber sequences are cofiber sequences.

Thm: If C is a stable ∞-category, then $H^0(C)$ is triangulated category.

Def: A dg category is a category enriched over k-mod. (Chain complexes)
- Given dg cat A, $H^0(A)$: homotopy category (H^0 on Hom)

Ex: A-mod $\xrightarrow{H^0}$ $K(A)$.

Def: A map $A \to B$ is quasi-iso. if $H^0(A) \xrightarrow{\simeq} H^0(B)$.

A-B bimodules are functors

$A^{op} \otimes B \longrightarrow \mathrm{Perf}_k$.

Note: $H^0(A)$ maybe not triangulated, need for "triangulate" + idem. complete it.

- X $A^{op}\otimes B$-mod $\rightsquigarrow$ functor $A^{mod} \to B^{mod}$.

- An A-B bimod X is Morita equivalence if it induces an equivalence on homotopy categories.

Rmk/ Additive invariants of dg categories (1) invariant Morita equivalence, whether it (2) preserve filtered colimits. (3 analogy of Waldhausen condition.

Thm [Blumberg-G - Tabuada] Connective alg K-theory is universal add inv.

Prop There is a model structure where weak equivalences are quasi-equiv or Morita equivalence.

Problem: These model structures are not compatible with internal Hom op.

(e.g. in 2006 Toen solves this problem)

- Assume k field or A is k-flat. ($\mathrm{Hom}_{\mathcal{M}}(X,Y)\otimes -$ preserve quasi-eq.)

Cliff - model structure

$\mathrm{rep}(A,B) = \{ X \in A\text{-}B \text{ bimod} \mid X \text{ induces a map from } H^0(A) \to H^0(B) \}$ full.

$(A \hookrightarrow A\text{-mod})$ Yoneda.

Thm This is internal hom in dg-cat $[qis^{-1}]$.

Thm $= \mathrm{Hom}_{A_\infty}(A,B)$.

Thm S set of morphisms in A. There exists $A \to A[S^{-1}]$ (via homotopy colimit

$\mathrm{colim}\Big(\coprod_{SES} (\cdot \to \cdot) \to A \Big)$

target

k-mod

Thm (Drinfeld by quot) $\mathcal{N} \subseteq \mathcal{A}$ full subcategory. $\exists\ Q : \mathcal{A} \to \mathcal{A}/\mathcal{N}$

Heuristically, for each $n \in \mathcal{N}$, a fixed truly separated map $u : n \xrightarrow{\pi} n$, with $d_u = id_n$.

Monta model structure: restricted to Monta fibrant dg categories.

Def: fibrant: pretriangulated dg category $\mathcal{A}$ such that $H^0(\mathcal{A})$ is idempotent-complete.

$\mathcal{A} \to \mathrm{Perf}(\mathcal{A})$ = pretriangulated $(\mathcal{A})$ - full ~~ss~~'s subcat generated by $\mathcal{A} \subseteq \mathcal{A}$-mod.

Thm [K, TT] Suppose $\mathcal{A} \xrightarrow{I} \mathcal{B} \xrightarrow{P} \mathcal{C}$ such that $PI = 0$.

Then, TFAE: (i) I is a $\ker(P)$, $P = \mathrm{coker}(I)$,

(ii) I ~~tato~~: $\mathrm{Perf}(\mathcal{A}) \xrightarrow{\simeq}$ thick subcat of $\mathrm{Perf}(\mathcal{B})$

$P : \mathrm{idem}(\mathcal{B}/\mathcal{A}) \xrightarrow{\simeq} \mathrm{Perf}(\mathcal{C})$

~~(iii) $P : D(\mathcal{B})/D(\mathcal{A}) \xrightarrow{\simeq} \mathcal{C}$~~

(iii) $D(\mathcal{A}) \xrightarrow{\simeq}$ thick subcat of $D(\mathcal{B})$

$D(\mathcal{A})/D(\mathcal{B}) \xrightarrow{\simeq} D(\mathcal{C})$.

Ex $X \xrightarrow{i} Y$ $Z = X \cup Y$.

$\mathrm{Perf}_Z(X) \to \mathrm{Perf}(X) \to \mathrm{Perf}(Y)$ satisfies conditions.

Thm [Cohn] Let $\mathbb{Q} \subset \mathrm{char}(k) = 0$. Then

$$(\text{dg-cat}_k)[\text{Morita eq}^{-1}] \xrightarrow{\simeq} \left(\begin{array}{c}Hk\text{-linear stable } \infty\text{-category} \\ \text{strongly compactly}\end{array}\right).$$

Hk = Eilenberg-Mac Lane spectrum of k.

Thm (Shipley) $H\mathbb{Z}$-mod $\simeq$ $\mathbb{Z}$-chain. (or for k ??)